

Name
Date
Course

Inverse Functions and Writing Equations Given Their Graphs (Homework)

Write an equation corresponding to each graph.

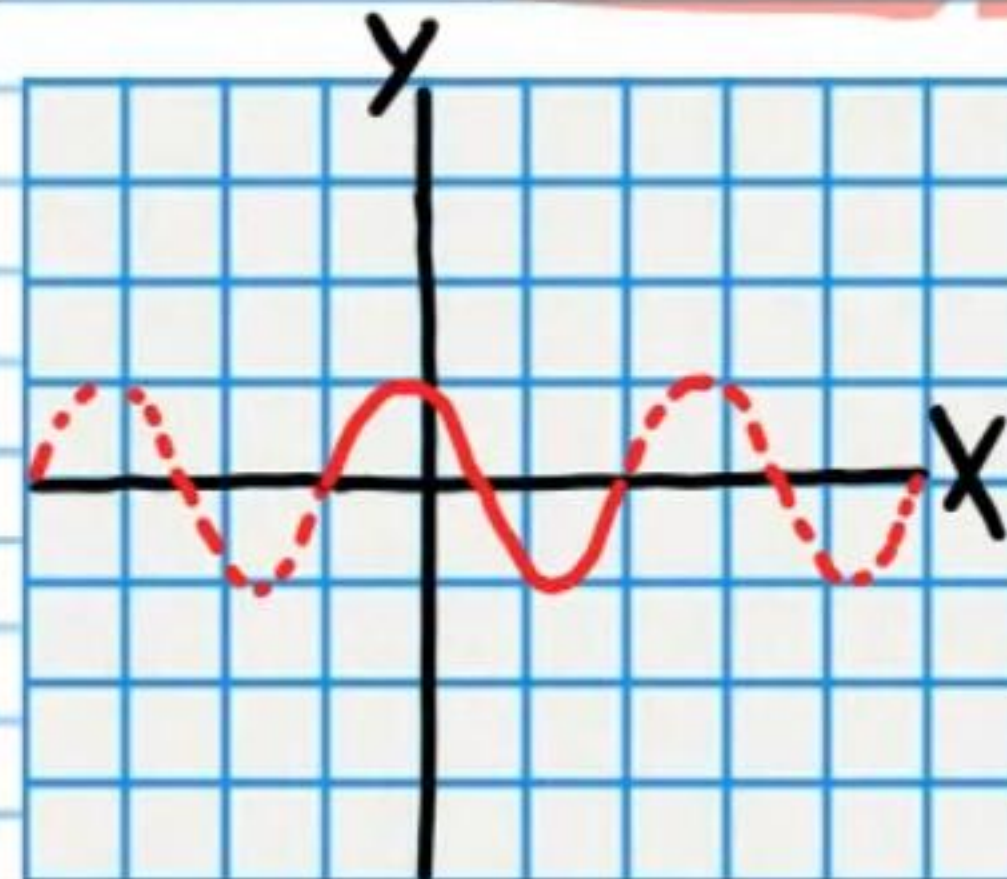
★ There is more than one equation that can be written for each graph (e.g. cosine functions or sine functions can represent the same graph). To avoid confusion, use the sine or cosecant functions rather than the cosine or secant functions. ★ Each horizontal unit on the graphs is 90° . Each vertical unit is one.

General Form of the Sine Function: $y = A \sin(\omega x - \phi) + B$

- ① Midline Equation:

Amplitude:

Omega:



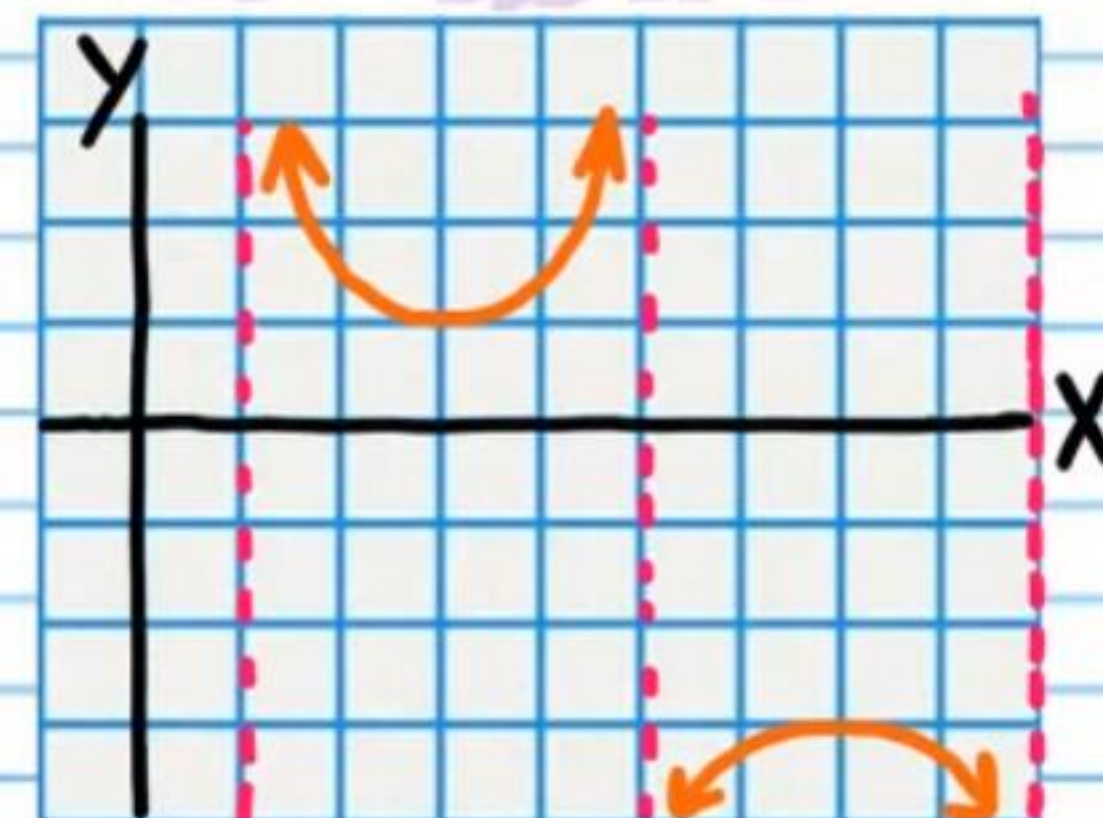
Phase Shift:

General Form of the Cosecant Function: $y = A \csc(\omega x - \phi) + B$

- ② Midline Equation:

A:

Omega:



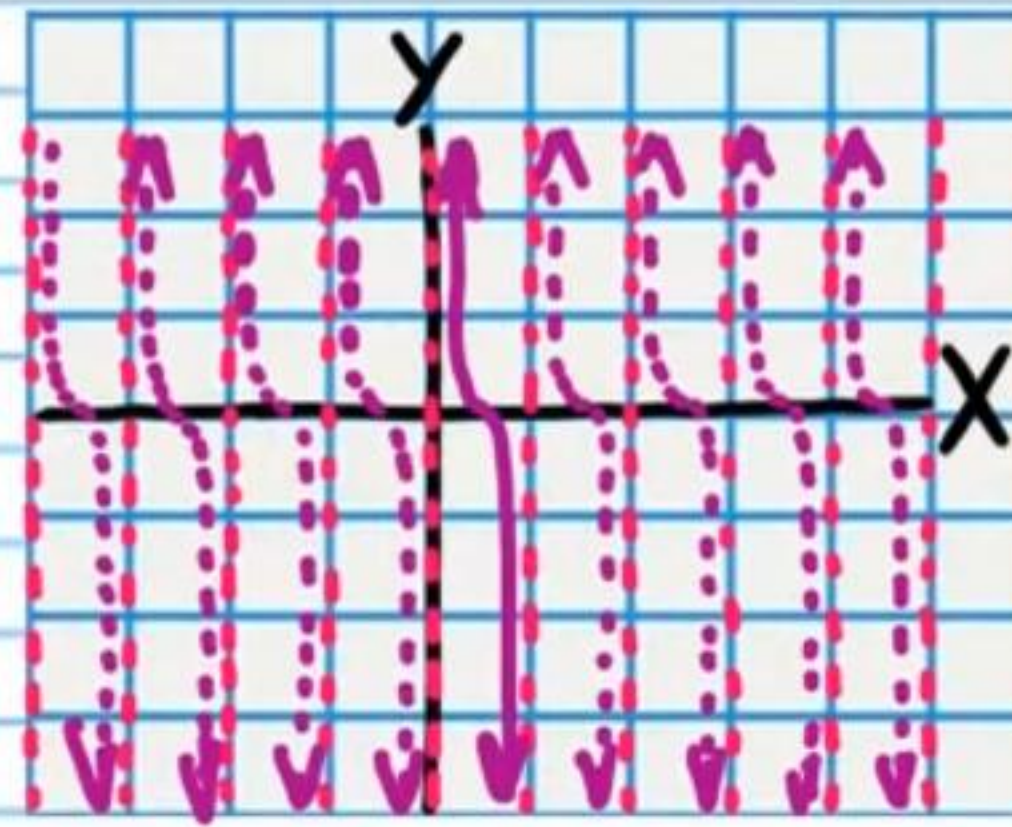
Phase Shift:

General Form of the Cotangent Function: $y = A \cot(Wx - \phi) + B$

③ Midline Equation:

A: ★ Assume $A=1$

Omega:



Phase Shift:

Find the values of the following expressions. You will need a calculator.

④ $\sin^{-1}(-0.87)$

⑤ $\tan^{-1} 3$

Find the values of the following expressions without a calculator.

⑥ $\cos^{-1}(\frac{1}{\sqrt{2}})$

⑦ $\tan^{-1}(\sqrt{3})$

⑧ $\sin^{-1}(-\frac{1}{2})$

Your answer will be negative.

⑨ $\csc^{-1}(\frac{2}{\sqrt{3}})$

⑩ $\sec^{-1}(1)$

⑪ $\cot^{-1}(0)$

★ ⑫ $\cos^{-1}(2)$

Homework Answers

① $y = \sin\left(\frac{4}{3}x + 120^\circ\right)$

② $y = 2\csc\left(\frac{1}{2}x - 45^\circ\right) - 1$

③ $y = \cot(2x)$

④ -60.46°

⑤ 71.57°

⑥ 45°

⑦ 60°

⑧ -30°

⑨ 60°

⑩ 0°

⑪ 90°

⑫ Undefined